

Computational Data checklist (de la Fuente Lab)

Please carefully go over the checklist below to ensure all data, software, and/or code aspects supporting the results reported in the manuscript comply with the lab's policies to ensure reproducibility and transparency. Please answer all the questions below and provide specific comments for each question you replied 'No' to. This checklist will need to be completed prior to submission of any paper from the lab that includes computational work.

1. Raw Data	Yes	No	
<i>All databases, websites, servers, and other sources used for predictions or analyses must be uploaded to Box.</i>	☐	☒	
<i>1b. Files must be labeled as Raw Data and include a brief description access date, version, number of entries, and any other relevant details.</i>	☐	☒	
2. Data sources	Yes	No	N/A
<i>1a. Are all data sources listed and publicly available?</i>	☒	☐	☐
<i>1b. If using an external database, is an access date or version number provided?</i>	☐	☒	☐
<i>1c. Are any potential biases in the source dataset reported and/or mitigated?</i>	☒	☐	☐
3. Data cleaning	Yes	No	N/A
<i>2a. Are the data cleaning steps clearly and fully described, either in text or as a code pipeline?</i>	☒	☐	☐
<i>2b. Is an evaluation of the amount of removed source data presented?</i>	☒	☐	☐
<i>2c. Are instances of combining data from multiple sources clearly identified, and potential issues mitigated?</i>	☒	☐	☐
4. Data representations	Yes	No	N/A
<i>3a. Are methods for representing data as features or descriptors clearly articulated, ideally with software implementations?</i>	☒	☐	☐
<i>3b. Are comparisons against standard feature sets provided?</i>	☒	☐	☐
5. Model choice	Yes	No	N/A
<i>4a. Is a software implementation of the model provided such that it can be trained and tested with new data?</i>	☐	☒	☐
<i>4b. Are baseline comparisons to simple/trivial models (for example, 1-nearest neighbour, random forest, most frequent class) provided?</i>	☒	☐	☐
<i>4c. Are baseline comparisons to current state-of-the-art provided?</i>	☒	☐	☐
6. Model training and validation	Yes	No	N/A
<i>5a. Does the model clearly split data into different sets for training (model selection), validation (hyperparameter optimization), and testing (final evaluation)? Are all cross-validation analysis details provided?</i>	☒	☐	☐
<i>5b. Is the method of data splitting (for example, random, cluster- or time-based splitting, forward cross-validation) clearly stated? Does it mimic anticipated real-world application?</i>	☒	☐	☐
<i>5c. Does the data splitting procedure avoid data leakage (for example, is the same composition present in the training and test sets)?</i>	☒	☐	☐
7. Code and reproducibility	Yes	No	N/A
<i>6a. Is the code or workflow available in a public repository?</i>	☐	☒	☐
<i>6b. Are scripts to reproduce the findings in the paper provided?</i>	☐	☒	☐
<i>6c. Have the authors clearly specified which versions of the software libraries they depend upon were used in the course of the work?</i>	☐	☒	☐

Checklist Comments for 'No' Answers:

Raw Data (Box) & 1b: Raw source files (Zenodo antibiogram, BV-BRC accession lists, intermediate manifests, RGI outputs) will be uploaded to the lab Box, labeled with description, access date, version, and entry counts.

1b (Data sources): Access dates for the Zenodo antibiogram repository and the BV-BRC genome retrieval will be added to the Methods, together with the Kleborate version used.

4a: The complete analysis code and reproduction scripts are being deposited in a public GitHub repository, to be linked in the manuscript before submission.

4c (Optional note): baselines evaluados bajo random split; comparación bajo ST-blocking queda como trabajo futuro.

5a: All cross-validation details are provided (5-fold StratifiedGroupKFold, random_state=4350); early stopping and hyperparameter selection were performed on internal validation splits within each training fold.

6a & 6b: The complete analysis code and reproduction scripts are being deposited in a public GitHub repository, to be linked in the manuscript before submission.

6c: A complete dependency list (requirements.txt) accompanies the repository; the manuscript specifies Python 3.12, Matplotlib 3.10.1, Seaborn 0.13.2, RGI 6.0.3, CARD 4.0.122.

Reviewer / Author Response Comments:

Inline comment 1 — *"please add all our affiliations, you can check one of our recent papers. thanks"*

Response: Done. The complete affiliation list is now provided, including the Machine Biology Group (Departments of Psychiatry and Microbiology), the Institute for Biomedical Informatics, the Institute for Translational Medicine and Therapeutics, the Departments of Bioengineering and Chemical and Biomolecular Engineering, the Department of Chemistry, and the Penn Institute for Computational Science.

Location: Title page, author affiliations.

Inline comment 2 — *"please input all citations"*

Response: Done. The sentence now cites references 1, 2, 3, 7–10, and 22. The reference list was expanded from 19 to 28 entries, adding previously missing tool and method citations (Kleborate, BV-BRC, DeLong, Clopper-Pearson, BLAST, CLSI M100), and all in-text citations were renumbered accordingly.

Location: Introduction, paragraph 2; References.

Inline comment 3 — *"please specify for all figures which software was used to make the figures"*

Response: Done. The Methods now state that all figures were generated in Python (v3.12) using Matplotlib (v3.10.1) and Seaborn v0.13.2

Location: Methods, opening line.

Inline comment 4 — *"specify (marginal comment on the validation table)"*

Response: Done. Table 1 now specifies, for each row: validation scheme, AUC with DeLong 95% confidence intervals, per-fold mean $\pm$ SD, and the role of each estimate (leakage comparator / primary result / ablation). The cohorts and prediction vectors are stated in the text (legacy held-out set, $n = 770$; pooled out-of-fold vector, $n = 3,702$).

Location: Results, Table 1; Methods, Train/validation split.

Critical Comment 1 — *"Reconstruct and document the complete cohort."*

Response: The Data source section now documents a step-by-step curation workflow with explicit numbers: 34,773 raw BV-BRC genomes → 4,869 isolates with explicit ertapenem labels (unknown/ambiguous discarded, no manual AST harmonization, stated as a limitation) → merged with the 684 resolved Zenodo isolates (238 unique; 446 already present) → 5,107 → anti-leakage exclusion of 1,384 genomes (exact duplicates and contradictory sequence isolates, defined as identical accessions with conflicting AST labels) plus 15 RGI failures → 3,708 master cohort → 3,702 OOF vector (6 isolates lost, acknowledged). The earlier 3,848-isolate cohort is explicitly identified as a legacy random-split benchmark from a different analysis universe, with the 140-isolate difference acknowledged as not reconstructable. A per-isolate manifest (accession, phenotype, ST, fold assignment, out-of-fold predictions) is provided in the analysis package. The duplicate audit was accession-level; a full near-identical sequence-similarity audit was not performed and is acknowledged as a limitation. Note: The complete public GitHub repository containing the analysis package will be uploaded before submission.

Location: Methods, Data source and genome retrieval; analysis package (GitHub).

Critical Comment 2 — *"Reconcile all sample-size and denominator inconsistencies."*

Response: All denominators now reconcile. Table 1 reports metrics and n per scheme; fold sizes sum to 3,702; the unknown-ST discrepancy (451 vs. 500) was audited and traced to 50 EuSCAPE orphan isolates minus 1 non-propagated isolate. MIL VME rates were recomputed from the OOF vector (MIL-only 34.1%, 95% CI 31.7–36.6; MIL+XGBoost 53.6%, 95% CI 51.0–56.2; Clopper-Pearson). Table 5 was rebuilt to reflect exclusively the rule baseline (0/1,383 carbapenemase-positive; 66/66 non-carbapenemase); the threshold sweep is explicitly labeled as the legacy random-split cohort; major-error rates and specificities are now reported for all models (rule 36.1% ME / 63.9%; MIL-only 20.4% / 79.6%; MIL+XGBoost 12.3% / 87.7%).

Location: Methods, Train/validation split; Results, ST-blocked validation performance and False negative profiling; Tables 1 and 5.

Critical Comment 3 — *"Make the random-versus-ST-blocked comparison architecture matched."*

Response: The manuscript now states explicitly that the random-split (hybrid, 0.9621) and ST-blocked primary (MIL-only, 0.8111) estimates use different configurations, and that their difference should not be interpreted as an architecture-matched estimate of clonal-structure effects. Architecture-matched differences are now reported: MIL-only 0.9182 → 0.8111 (11.6% relative decrease); hybrid 0.9621 → 0.7981 (17.0% relative decrease). Both schemes used identical preprocessing and hyperparameters. Early stopping and hyperparameter selection were performed on internal validation splits within each training fold.

Location: Abstract; Results, Table 1 and following paragraph; Discussion; Conclusion.